



2025

J22

BOOKLET NO.

138091

**Civil Engineering**  
**Paper – I**

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

**Question Paper Specific Instructions**

*Please read each of the following instructions carefully before attempting questions :*

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

**Note :** Candidates will be allowed to use Scientific (Non-programmable type) Calculators.

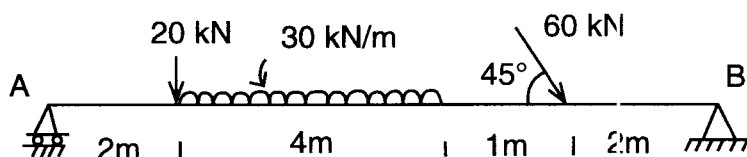
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## SECTION – A

**Q1.** Answer the following questions in about **150** words. **(5×10=50)**

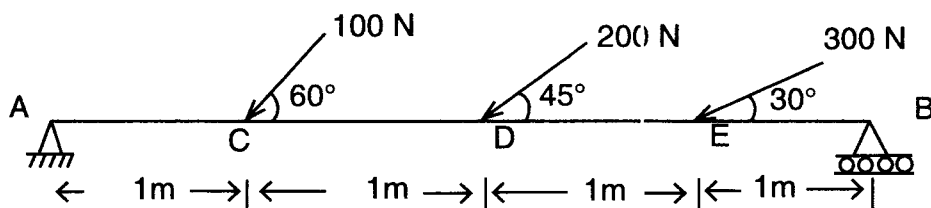
- Q1. (a)** Explain law of transmissibility of forces. Find the reactions at supports A and B of the loaded beam shown in Fig. **10**



- Q1. (b)** What are lacings and battens in a built-up column ? Explain their functions. Explain the concept of shear lag in an angle tension member connected by only one leg and how it affects design strength calculation. **10**

- Q1. (c)** What do you mean by point of contra flexure ? Is point of contra flexure and point of inflexion are different ?

A horizontal beam AB of length 4 m is hinged at A and supported on rollers at B. The beam carries inclined loads of 100 N, 200 N and 300 N inclined at  $60^\circ$ ,  $45^\circ$  and  $30^\circ$  to horizontal as shown in Fig. Draw shear force and bending moment diagram for the beam. **10**





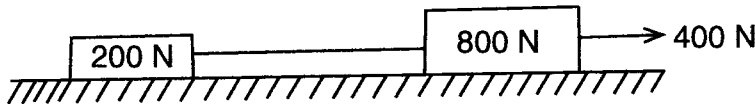
- Q1. (d)** What is bolt ? State types of bolts and explain high strength bolts.

A single riveted double cover butt joint is used to connect two plates 18 mm thick with chain riveting. The rivets used are power driven 20 mm in diameter at a pitch of 60 mm. Find out pitch load per pitch length.

Assume –  $\tau_{vf} = 100 \text{ N/mm}^2$ ,  $\sigma_{pf} = 300 \text{ N/mm}^2$ ,  $f_y = 250 \text{ MPa}$  **10**

- Q1. (e)** Find the collapse load in a fixed beam of span  $L$  carrying uniformly distributed load over the entire span. Plastic moment capacity is  $M_p$  throughout. **10**

- Q2. (a)** What is moment of inertia ? Derive mathematical expression for moment of inertia of a semi circle about diametral axis. Two weights 800 N and 200 N are connected by a thread and they move along rough horizontal plane under the action of a force of 400 N applied to the 800 N weight as shown in Fig. The coefficient of friction between the sliding surface of the weights and the plane is 0.3. Using D'Alembert's principle determine acceleration of the weight and tension in thread. **20**



- Q2. (b)** What is workability of concrete ? Which factors influence workability of concrete ? Explain methods of proportioning concrete mixes. **20**
- Q2. (c)** Explain the term elastic centre and its significance in the analysis of fixed arches. **10**



- Q3. (a)** Describe slope and deflection of a simply supported beam carrying a point load at the centre by Mohr's theorem.
- A beam of length 5 m and of uniform rectangular section is supported at its ends and carries uniformly distributed load over the entire length. Calculate depth of the section if maximum permissible bending stress is  $8 \text{ N/mm}^2$  and central deflection not to exceed 10 mm. Take value of  $E = 1.2 \times 10^4 \text{ N/mm}^2$ . **20**
- Q3. (b)** What is moment of resistance ? Explain in brief balanced, under reinforced and over reinforced sections. **20**
- Q3. (c)** State types of prestressing and describe losses of prestress. **10**
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- Q4. (a)** What do you understand by flexibility matrix and stiffness ? Prove that these are inverse of each other. **20**
- Q4. (b)** Discuss in brief design requirements of circular water tank resting on ground. **20**
- Q4. (c)** Write a note on displacement methods of analysis of indeterminate beams. **10**
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### SECTION – B

- Q5.** Answer the following questions in about 150 words. **(5×10=50)**
- Q5. (a)** State properties of fluids. The vertical side of a reservoir has a rectangular opening 2.5 m long and 1.2 m high. It is closed by using 4 bolts placed at corner of the opening. What would be the tension in bolts, if water stands to a height of 1.8 m above the top edge of the opening which is horizontal ? **10**
- Q5. (b)** What are the soil consistency limits ? A fully saturated clay specimen has a water content of 32% and mass specific gravity of 2.1. When the specimen is oven dried, its mass specific gravity reduces to 1.65. Assuming that the void ratio remains unchanged during drying and the specimen does not undergo any volume change. Determine the specific gravity of clay solids. **10**
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- Q5. (c)** State methods of dimensional analysis and outline procedure of Buckingham's  $\pi$  theorem. **10**
- Q5. (d)** Define consolidation. Determine the average coefficient of permeability in horizontal and vertical directions for a deposit consisting of three layers. The first layer has a thickness of 8 m and coefficient of permeability  $4 \times 10^{-3}$  mm/sec. the second layer has thickness of 3 m and coefficient of permeability  $2 \times 10^{-5}$  mm/sec and third layer has a thickness of 5 m and coefficient of permeability  $6 \times 10^{-2}$  mm/sec. Assume layers are isotropic. **10**
- Q5. (e)** Derive expression for mean velocity for laminar flow between parallel plates. **10**
- For laminar flow of an oil having dynamic viscosity  $\mu = 1.766$  pa.s in a 0.3m of diameter pipe, the velocity distribution is parabolic with maximum velocity of 3 m/s at the centre of pipe.
- Calculate the shearing stresses at the pipe wall and with in the fluid 50mm from the pipe wall.
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- Q6. (a)** What do you mean by equipotential line and line of constant stream function ? **20**  
Describe use and limitation of flow net.  
An open circular cylinder of 15 cm diameter and 100cm long contains water upto a height of 80 cm. Find the maximum speed at which cylinder is to be rotated about its vertical axis so that no water spills.
- Q6. (b)** A normally consolidated clay layer of thickness 6 m is sandwiched between a pervious sand layer at the top and an impervious rock layer at bottom. The clay has following properties.  
Initial void ratio  $e_0 = 0.95$   
Compression index  $e_c = 0.38$
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Coefficient of consolidation =  $C_v = 3 \times 10^{-3} \text{ cm}^2/\text{s}$

Saturated unit weight of clay  $\gamma_{\text{sat}} = 18.5 \text{ kN/m}^3$

The water table is at ground level. A uniform load  $120 \text{ kN/m}^2$  is applied at the ground surface over a large area. Determine (i) Ultimate consolidation settlement of clay layer (ii) Time required for 70% consolidation.

Assume one dimensional consolidation and use  $T_v = 0.405$

(for 70% consolidation)

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- Q6. (c)** Two parallel plates kept 0.1 m apart have laminar flow of oil between them with a maximum velocity of 1.5 m/s. Calculate discharge per metre width, the Shear stress at the plates, the difference in pressure in pascals between two points 20 m apart. The velocity gradient at the plates and velocity at 0.02 m from the plate. Take viscosity of oil to be  $2.453 \text{ N.S/m}^2$ .

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- Q7. (a)** Explain characteristics of laminar and turbulent boundary layer. What factors are affecting boundary layer thickness ?

A rough pipe line of diameter 0.1 m carries water at  $20^\circ\text{C}$  at the rate of 50 l/s. If average height of the protrusions on the pipe surface is 0.15 mm. Calculate the friction factor, maximum velocity, shear stress at the pipe surface and shear velocity. For water at  $20^\circ\text{C}$  take  $\gamma = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$  and  $\rho = 1000 \text{ kg/m}^3$ .

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- Q7. (b)** A rectangular footing of size  $3.0 \text{ m} \times 2.0 \text{ m}$  is founded at a depth of 2 m below ground level in  $C - \phi$  soil. The soil properties are cohesion  $C = 20 \text{ kN/m}^2$ . Angle of internal friction  $\phi = 28^\circ$ , unit weight  $\gamma = 18 \text{ kN/m}^3$ , saturated unit weight  $\gamma_{\text{sat}} = 9.81 \text{ kN/m}^3$ . The water table is located 1 m below ground level. Use Terzaghi's bearing capacity theory with shape factors and determine

1) Net ultimate bearing capacity

2) Net safe bearing capacity. Assume factor of safety of 3. Bearing capacity factors for  $\phi = 28^\circ$   $N_c = 25$ ,  $N_q = 14.7$ ,  $N_\gamma = 16$  and shape factor  $S_c = 1.3$ ,

$S_q = 1.2$ ,  $N_\gamma = 0.8$

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- Q7. (c)** Explain principles of the design of retaining walls.

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- Q8.** (a) State the conditions under which the rectangular section of an open channel will be most economical ? Derive these conditions. The depth of flow of water at a certain section of rectangular channel of 2 m wide is 0.3 m. The discharge through the channel is  $1.5 \text{ m}^3/\text{s}$ . Determine whether a hydraulic jump will occur and if so find its height and loss of energy per kg of water. **20**
- Q8.** (b) Explain the method for the design of raft by elastic method. **20**
- Q8.** (c) What is turbine ? How will you classify turbines ? Explain main characteristics of turbines. **10**
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